

Finite-particle singular realization and heat passage

HOCF–RIESZ, Round 006

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Scope and status. This memorandum records the finite- N assertion THM–026. Its complete construction, fresh reconstruction and hostile review have separate immutable Markdown reports; subsequent dispositions are in the canonical theorem ledger. The result concerns particle realization and law domination. It does not supply the domain of a singular pair corrector or estimates uniform in particle number.

Model and precise conclusions

The torus has Haar mass one and characters $e^{2\pi i k \cdot x}$. Fix $d \geq 3$, $0 < s \leq d-2$, $N \geq 2$, $0 \leq \nu < \infty$, a finite horizon T , and a smooth real periodic potential V . Set $b = -\nabla V$, $K = -\nabla g_s$, where

$$\widehat{g}_s(0) = 0, \quad \widehat{g}_s(k) = \pi^{s-d/2} \frac{\Gamma((d-s)/2)}{\Gamma(s/2)} |k|^{s-d}, \quad k \neq 0.$$

The previously proved heat representation gives $g_s(z) = |z|^{-s} + h_s(z)$ with a smooth even local remainder, $g_s, K \in L^1$, and

$$D := \operatorname{div} K = \begin{cases} s(d-2-s)g_{s+2} dz, & s < d-2, \\ c_d(\delta_0 - dz), & s = d-2, \end{cases} \quad c_d = (d-2)|\mathbb{S}^{d-1}|.$$

Choose $a, \kappa \geq 0$ with $\operatorname{div} b \geq -a$ and $D \geq -\kappa dz$, and put $C_N = Na + (N-1)\kappa$. With independent standard Brownian motions, the equation is

$$dX_i = \left[b(X_i) + \frac{1}{N} \sum_{j \neq i} K(X_i - X_j) \right] dt + \sqrt{2\nu} dW_i.$$

From every fixed distinct configuration it has a unique global collision-free pathwise realization. The path and transition kernels are jointly Borel and have the deterministic-time conditional Markov property. Each fixed realized finite horizon has a positive minimum pair distance. No single exceptional set for all uncountably many starts is asserted.

For the same initial state and Brownian motions, replacing K by its positive heat convolution K_ε gives

$$\sup_{0 \leq t \leq T} \max_i \operatorname{dist}_{\mathbb{T}^d}(X_i^\varepsilon(t), X_i(t)) \longrightarrow 0 \quad \text{almost surely as } \varepsilon \downarrow 0.$$

This is a fixed-parameter full-family limit, with no uniform rate. An initial probability density $F_0 \in L^\infty((\mathbb{T}^d)^N)$, independent of the Brownian motions, evolves to a density with

$$F_t \leq \|F_0\|_\infty e^{C_N t} \quad \text{almost everywhere.}$$

For iid initial density bounded by M , the factor is $M^N e^{C_N t}$. The estimate does not propagate independence or give a uniform bound in N .

Energy controls every partial collision

Write $v_* = \min V$, $g_* = \inf_{z \neq 0} g_s(z) > -\infty$, and define

$$H_N = \sum_i V(x_i) + \frac{1}{N} \sum_{i < j} g_s(x_i - x_j), \quad \mathcal{E}_N = H_N - Nv_* - \frac{N-1}{2} g_*.$$

Each summand in

$$\mathcal{E}_N = \sum_i (V(x_i) - v_*) + \frac{1}{N} \sum_{i < j} (g_s(x_i - x_j) - g_*)$$

is nonnegative. Thus a single colliding pair forces the energy to infinity, including at simultaneous clusters or disjoint partial collisions. All energy sublevels are compact in the collision-free configuration space. If $\delta = \min_{i < j} \text{dist}(x_i, x_j) \leq r_0 = 1/4$, a fixed $A > 0$ determined by the local remainder gives

$$\mathcal{E}_N \geq \frac{\delta^{-s} - A}{N}.$$

The complete drift B is $-\nabla H_N$ and the full trace is

$$\Delta_{Nd} H_N = \sum_i \Delta V(x_i) + \frac{2}{N} \sum_{i < j} \Delta g_s(x_i - x_j) \leq C_N.$$

The coefficient is $2/N$. At Coulomb, the punctured periodic Laplacian is $\Delta g_s = c_d$; it is not the zero Euclidean principal Laplacian. For $Q_N = C_N - \Delta H_N \geq 0$, the exact generator identity is

$$L_N \mathcal{E}_N = \nu C_N - |B|^2 - \nu Q_N.$$

The square is the full squared total drift. Its triple cross terms can have either sign and are not discarded. Three equally spaced collinear particles have zero leading force on the middle particle but divergent total pair energy; no individual-force coercivity is needed.

Construction, stopping and global integrability

Smooth periodic force cutoffs agree with K away from a shrinking collision tube. Subtracting the continuous additive driving path gives a globally Lipschitz integral equation; Picard iteration and Gronwall construct its unique solution and Borel dependence. Local uniqueness patches these solutions up to collision. On the torus the only possible finite lifetime approaches that set.

Let σ_R be the first energy-level exit. Before this stop the coefficients and energy derivatives lie in a fixed compact smooth region. Smooth Itô calculus therefore gives

$$\begin{aligned} \mathcal{E}_N(X_{t \wedge \sigma_R}) + \int_0^{t \wedge \sigma_R} |B|^2 dr + \nu \int_0^{t \wedge \sigma_R} Q_N dr &= \mathcal{E}_N(x) + \nu C_N(t \wedge \sigma_R) + M_R(t), \\ [M_R]_t &= 2\nu \int_0^{t \wedge \sigma_R} |B|^2 dr. \end{aligned}$$

The stopped integrand is bounded, so M_R is a true square-integrable martingale. Taking expectations, using nonnegativity, yields

$$\mathbb{P}_x(\sigma_R \leq T) \leq \frac{\mathcal{E}_N(x) + \nu C_N T}{R}.$$

Energy coercivity implies that these stops exhaust the local lifetime. Letting R tend to infinity proves noncollision and global uniqueness, before using positive minimum path separation.

For independent initial data with $J_0 = \mathbb{E} \mathcal{E}_N(X_0) < \infty$, Fatou first proves integrability of $\int_0^T |B|^2$ and $\int_0^T \nu Q_N$. The unstopped energy martingale is then square integrable. Its omitted bracket has expectation

tending to zero, so the stopped martingales converge in L^2 . Passing the localized identity and taking expectations now gives the equality

$$\mathbb{E}\mathcal{E}_N(X_t) + \mathbb{E} \int_0^t |B|^2 dr + \mathbb{E} \int_0^t \nu Q_N dr = J_0 + \nu C_N t.$$

At $\nu = 0$ the last integral is identically zero; no division by ν occurs. The same stopped argument at a distance threshold ρ gives

$$\mathbb{P} \left(\min_{[0,T]} \delta(X_t) \leq \rho \right) \leq \frac{N(J_0 + \nu C_N T)}{\rho^{-s} - A} \leq 2N(J_0 + \nu C_N T) \rho^s$$

when $0 < \rho \leq r_0$ and $\rho^{-s} \geq 2A$. For bounded F_0 , the required energy mean is finite because

$$\int \mathcal{E}_N dX = N \left(\int V - v_* \right) - \frac{N-1}{2} g_* < \infty.$$

This initial integrability is established before any density propagation.

The countable cutoff construction yields a Borel set of good start/path pairs, whose section has probability one at each distinct start. Independent Brownian increments and Fubini at a random current state justify deterministic-time restart and the Markov property. A universal exceptional set is unnecessary.

Heat coupling and density passage

On every compact set away from zero, $K_\varepsilon \rightarrow K$ in C^1 . To prove this, separate a smooth part agreeing with K near that set from an L^1 distant part. Gaussian derivative tails bound the latter by an inverse power of ε times $e^{-c/\varepsilon}$. Fix a noncolliding path and its positive minimum separation. The Brownian terms cancel in the same-noise difference equation. A stopped Gronwall bound prevents the smooth trajectory from leaving a small neighborhood of that path for every sufficiently small ε . This proves the full-family uniform path limit.

The regularized full drift has exact divergence

$$\operatorname{div}_{Nd} B^\varepsilon = \sum_i \operatorname{div} b(x_i) + \frac{2}{N} \sum_{i < j} D_\varepsilon(x_i - x_j) \geq -C_N.$$

For each continuous driving path its smooth flow is a spatial diffeomorphism: solve the translated integral equation backwards for its inverse. Its ordinary variational equation gives a Jacobian determinant at least $e^{-C_N t}$. Change of variables, the bound on F_0 , and Brownian expectation give

$$\mathbb{E}G(X_t^\varepsilon) \leq \|F_0\|_\infty e^{C_N t} \int G dX \quad (G \geq 0 \text{ continuous}).$$

The same-noise convergence and bounded convergence first pass this inequality for continuous tests. Increasing continuous approximations of open-set indicators, followed by Haar outer regularity, extend it to all Borel sets. This proves the asserted density domination without assuming a singular Fokker–Planck solution, convergence for all Borel terminal tests, or an inverse for the limiting singular flow.

The remaining corrector-domain line

These results do not justify applying Itô calculus to the bounded Borel full pair inverse from Round005. Its spatial and time derivatives, differentiated background contractions, singular generator identity, and residual/bracket approximation remain separate obligations. A finite- N density bound transfers an already proved integrable function; it cannot supply a missing derivative or turn a nonintegrable expression into an integrable one. Uniform evolved-law estimates and the critical hierarchy remain open, with the original energy-floor and microscopic regimes unchanged.